

数值分析第十六次作业

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1. 设 f, g 是以 2π 为周期的分片光滑函数, 定义

$$\hat{f}_k := \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx \in \mathbb{C}, \quad \forall k \in \mathbb{Z},$$

以及卷积

$$(f * g)(x) := \int_{-\pi}^{\pi} f(x-y)g(y) dy.$$

证明以下等式:

(1) 如果 $f \in C^1$, 则 $(f')_k^\wedge = ik \hat{f}_k$;

(2) $(f(x-a))_k^\wedge = e^{-ika} \hat{f}_k$;

(3) $(f * g)_k^\wedge = 2\pi \hat{f}_k \hat{g}_k$.

解答.

(1) 由周期性和分部积分可得

$$\begin{aligned} (f')_k^\wedge &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f'(x) e^{-ikx} dx \\ &= \frac{1}{2\pi} \left(f(x) e^{-ikx} \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} (-ik) f(x) e^{-ikx} dx \right) \\ &= ik \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx \\ &= ik \hat{f}_k. \end{aligned}$$

(2) 由周期性可得

$$\begin{aligned} (f(x-a))_k^\wedge &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-a) e^{-ikx} dx \\ &= \frac{1}{2\pi} \int_{-\pi-a}^{\pi-a} f(t) e^{-ik(t+a)} dt \\ &= e^{-ika} \cdot \frac{1}{2\pi} \int_{-\pi-a}^{\pi-a} f(t) e^{-ikt} dt \\ &= e^{-ika} \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-ikt} dt \end{aligned}$$

$$= e^{-ika} \hat{f}_k.$$

(3) 由周期性可得

$$\begin{aligned} (f * g)_k^\wedge &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (f * g)(x) e^{-ikx} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\int_{-\pi}^{\pi} f(x-y) g(y) dy \right) e^{-ikx} dx \\ &= \int_{-\pi}^{\pi} g(y) e^{-iky} \cdot \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-y) e^{-ik(x-y)} dx \right) dy \\ &= \int_{-\pi}^{\pi} g(y) e^{-iky} \cdot \left(\frac{1}{2\pi} \int_{-\pi-y}^{\pi-y} f(t) e^{-ikt} dt \right) dy \\ &= \int_{-\pi}^{\pi} g(y) e^{-iky} \cdot \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-ikt} dt \right) dy \\ &= \int_{-\pi}^{\pi} g(y) e^{-iky} \hat{f}_k dy \\ &= 2\pi \hat{f}_k \hat{g}_k. \end{aligned}$$

□

2. 设向量 $\mathbf{a} = (a_0, a_1, \dots, a_{N-1})^T$ 的离散 Fourier 变换为 $\hat{\mathbf{a}} = \mathbf{c} = (c_0, c_1, \dots, c_{N-1})^T$, 其中 a_k, c_k 对下标以 N 为周期. $\forall 0 \leq n \leq N-1$, 定义向量 $\mathbf{b}^{(n)}$ 和 $\mathbf{d}^{(n)}$, 其分量分别为

$$b_k^{(n)} = a_k \cos\left(\frac{2\pi kn}{N}\right), \quad d_k^{(n)} = c_{k+n} + c_{k-n}, \quad \forall 0 \leq k \leq N-1.$$

求证: $(\mathbf{b}^{(n)})^\wedge = \frac{1}{2} \mathbf{d}^{(n)}$.

解答.

$\forall 0 \leq k \leq N-1$, 由于

$$\begin{aligned} (b^{(n)})_k^\wedge &= \sum_{j=0}^{N-1} b_j^{(n)} e^{-jk \frac{2\pi i}{N}} = \sum_{j=0}^{N-1} a_j \cos\left(\frac{2\pi jn}{N}\right) e^{-jk \frac{2\pi i}{N}}, \\ \frac{1}{2} d_k^{(n)} &= \frac{1}{2} (c_{k+n} + c_{k-n}) = \frac{1}{2} \sum_{j=0}^{N-1} a_j \left(e^{-j(k+n) \frac{2\pi i}{N}} + e^{-j(k-n) \frac{2\pi i}{N}} \right) = \sum_{j=0}^{N-1} a_j \cos\left(\frac{2\pi jn}{N}\right) e^{-jk \frac{2\pi i}{N}}, \end{aligned}$$

故 $(b^{(n)})_k^\wedge = \frac{1}{2} d_k^{(n)}$, 从而 $(\mathbf{b}^{(n)})^\wedge = \frac{1}{2} \mathbf{d}^{(n)}$.

□

3. 设向量 $\mathbf{a} = (a_0, a_1, \dots, a_{N-1})^T$, 其中 a_k 对下标以 N 为周期, 定义向量 $\mathbf{b}$, 其分量为

$$b_k = a_{-k}, \quad \forall 0 \leq k \leq N-1.$$

求证: $\hat{\mathbf{a}} = N\mathbf{b}$.

解答.

$\forall 0 \leq k \leq N-1$, 由于

$$\hat{a}_k = \sum_{j=0}^{N-1} \hat{a}_j e^{-jk \frac{2\pi i}{N}} = \sum_{j=0}^{N-1} \sum_{\ell=0}^{N-1} a_\ell e^{-j\ell \frac{2\pi i}{N}} e^{-jk \frac{2\pi i}{N}} = \sum_{\ell=0}^{N-1} a_\ell \sum_{j=0}^{N-1} e^{-j(k+\ell) \frac{2\pi i}{N}} = \sum_{\ell=0}^{N-1} a_\ell \cdot N \delta_{k+\ell,0} = N a_{-k} = N b_k,$$

故 $\hat{\mathbf{a}} = N \mathbf{b}$.

□

4. 设向量 $\mathbf{a} = (a_0, a_1, \dots, a_{N-1})^T$, 其中 a_k 对下标以 N 为周期, 定义向量 $\mathbf{b}$ 和 $\mathbf{c}$, 其分量分别为

$$b_k = \hat{a}_{k+1} - \hat{a}_k, \quad c_k = \left(e^{-k \frac{2\pi i}{N}} - 1 \right) a_k, \quad \forall 0 \leq k \leq N-1,$$

求证: $\mathbf{b} = \hat{\mathbf{c}}$.

解答.

$\forall 0 \leq k \leq N-1$, 由于

$$\begin{aligned} b_k &= \hat{a}_{k+1} - \hat{a}_k = \sum_{j=0}^{N-1} a_j \left(e^{-j(k+1) \frac{2\pi i}{N}} - e^{-jk \frac{2\pi i}{N}} \right), \\ \hat{c}_k &= \sum_{j=0}^{N-1} c_j e^{-jk \frac{2\pi i}{N}} = \sum_{j=0}^{N-1} a_j \left(e^{-j \frac{2\pi i}{N}} - 1 \right) e^{-jk \frac{2\pi i}{N}} = \sum_{j=0}^{N-1} a_j \left(e^{-j(k+1) \frac{2\pi i}{N}} - e^{-jk \frac{2\pi i}{N}} \right), \end{aligned}$$

故 $b_k = \hat{c}_k$, 从而 $\mathbf{b} = \hat{\mathbf{c}}$.

□